



# Residual-based cointegration and non-cointegration tests for cointegrating polynomial regressions

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## Abstract

Cointegrating polynomial regressions (CPRs), i.e., regressions that include deterministic terms, integrated processes and powers of integrated processes as explanatory variables and stationary errors, have become prominent in several fields of applications, e.g., in the analysis of environmental Kuznets curves. A key issue, as always in cointegration analysis, is testing for the presence or absence of a cointegrating relationship. This paper discusses two complementary tests: one with the null hypothesis of cointegration and one with the null hypothesis of the absence of cointegration. It is shown that (inter alia) for the empirically most relevant case, in which only one of the integrated regressors occurs as regressor also with higher powers, critical values can be simulated and are provided for a variety of specifications. Finally, the usage of the tests is illustrated for the environmental Kuznets curve for carbon and sulfur dioxide emissions. The illustration also investigates the sensitivity of the test decisions with respect to kernel and bandwidth choices, sample size and data vintage.

**Keywords** Cointegrating polynomial regression · Cointegration · Unit root · Testing · Environmental Kuznets curve · Material Kuznets curve · Exchange rate target-zone

**JEL Classification** C12 · C13 · Q20

## 1 Introduction

Recent years have seen growing theoretical and empirical interest in estimating and testing for nonlinear cointegrating relationships. A very voluminous body of applied

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literature in this respect has been and continues to be generated within the environmental economics literature. The so-called environmental Kuznets curve (EKC) hypothesis, which postulates an inverted U-shaped relationship between the level of economic development, typically measured by (the logarithm of) GDP per capita, and (the logarithm of) pollution or emissions per capita is one of the most widely studied empirical relationships in environmental economics. Early survey papers like Stern (2004) and Yandle et al. (2004) already count more than one-hundred refereed publications. The term EKC refers by analogy to the inverted U-shaped relationship between the level of economic development and the degree of income inequality postulated by Kuznets (1955) in his 1954 presidential address to the American Economic Association. Estimation of an EKC entails, in its simplest parametric form, a regression of emissions on GDP and its square to allow for a U or inverted U shape. Given that the logarithm of GDP per capita is often found to be unit root non-stationary, this is thus a regression where as regressors a unit root process and its square are present.<sup>1</sup> In doing so, the empirical EKC literature in large parts fails to acknowledge the fact that if the logarithm of GDP per capita is a unit root process, the square of the logarithm of GDP per capita cannot be a unit root process and continues to use “standard” unit root and cointegration techniques. This means that a large part of the empirical EKC literature (see, e.g., Friedl and Getzner (2003), Perman and Stern (2003) or Galeotti et al. (2009) from a large and still growing list of such contributions) simply treats GDP and its square as two integrated processes and uses standard unit root and cointegration test and estimation methods.<sup>2</sup> This is, notwithstanding its popularity, a problematic practice since tailor-made methods designed for regressions involving integrated processes and nonlinear transformations thereof are available.<sup>3</sup>

Wagner and Hong (2016) develop estimation and inference theory for the special case of *nonlinear* cointegrating relationships including deterministic components as well as unit root processes and their powers as explanatory variables. They refer to such relationships as cointegrating polynomial regressions (CPRs). More precisely, Wagner and Hong (2016) extend the FM-OLS estimator of Phillips and Hansen (1990) from cointegrating linear to cointegrating polynomial regressions.<sup>4</sup> In addition to hypothesis

<sup>1</sup> The seminal empirical study of Grossman and Krueger (1993) uses a third-order polynomial in GDP, whereas the more popular quadratic specification has been initiated by Holtz-Eakin and Selden (1995).

<sup>2</sup> Another literature where this type of relationship occurs is the so-called intensity-of-use or material Kuznets curve (MKC) literature that investigates the potentially inverted U-shaped relationship between GDP and energy or metals use (see, e.g., Labson and Crompton 1993 or Grabarczyk et al. 2018). Almost by definition, the study of inequality, put to the center of attention again by Piketty (2014), when performed with time series data can make use of methods for cointegrating polynomial regressions. The same holds true for the exchange rate target-zone literature, see, e.g., Darvas (2008) or Svensson (1992).

<sup>3</sup> Bradford et al. (2005) and Wagner (2008, 2015) contain critical assessments of this practice. Stypka et al. (2019) derive and discuss the asymptotic properties of using methods, in particular FM-OLS, developed for cointegrating linear regressions in a cointegrating polynomial regression setting. The results of that paper show that “standard” cointegration and non-cointegration tests lead to asymptotically invalid inference, not least because of wrong critical values being used. This problem is overcome with the tests and *proper* critical values provided in this paper.

<sup>4</sup> Also, e.g., Chang et al. (2001), provide extensions of FM-OLS to nonlinear cointegration settings. While these authors consider more general functions than Wagner and Hong (2016), they assume serially uncorrelated errors. Wagner and Hong (2016) allow for serially correlated errors that are allowed to be dynamically correlated with the regressors, thereby considering the same setting as commonly used in linear cointegra-

testing with respect to the estimated parameters, they furthermore also investigate specification tests based on augmented or auxiliary regressions in detail. One important aspect that is not treated exhaustively in that paper is testing for cointegration and for the absence of cointegration, respectively. To be precise, only testing the null hypothesis of cointegration is considered in Wagner and Hong (2016, Propositions 5 and 6), whereas testing the null hypothesis of no cointegration is not considered. Furthermore, Wagner and Hong (2016) do not provide critical values. The present paper considers tests for both null hypotheses in detail and also provides critical values. We (re-)consider, with this test already discussed in Wagner and Hong (2016), in detail an extension of the Shin (1994) cointegration test, itself an extension of the KPSS stationarity test of Kwiatkowski et al. (1992), from cointegrating linear to cointegrating polynomial regressions. Furthermore, we consider an extension of a variance ratio test for the absence of cointegration of Phillips and Ouliaris (1990). The null hypothesis of the latter test is that the relationship is not cointegrating but spurious. It turns out that in the general case of several integrated regressors entering the CPR with powers higher than one, neither of the two tests has a nuisance-parameter-free limiting distribution that could be tabulated. This has already been observed for the KPSS-type test in Wagner and Hong (2016). Consequently, for the Shin- or KPSS-type test Wagner and Hong (2016) and, based on D-OLS estimation rather than FM-OLS estimation, Choi and Saikkonen (2010) consider sub-sampling versions of the KPSS-type test statistic that results in nuisance-parameter-free limiting null distributions, at the expense of a drastically reduced sample size due to sub-sampling. For the relatively small samples available for typical macroeconomic applications, a sub-sampling based test is potentially only of limited practical value. For the extension of the test of Phillips and Ouliaris (1990), this sub-sampling approach does not lead to asymptotically nuisance-parameter-free limiting distribution.

However, using the terminology of Vogelsang and Wagner (2016), in case of a *full design* CPR relationship it can be shown that the limiting distributions of both tests can be tabulated. Full design refers to a situation in which the limiting distribution of the FM-OLS estimator—and a fortiori the limiting partial sum processes of the residuals—can be expressed as a functional of standard Brownian motions rather than of non-standard Brownian motions. The empirically most relevant case, with only one of the integrated processes entering the CPR with powers higher than one is straightforwardly seen to be of full design. “Appendix B” of the paper provides tables with critical values for this case, for up to four integrated regressors, up to power four of the one integrated regressor entering with higher powers for the usual specifications of the deterministic component (no deterministic component, intercept only, intercept and linear trend). Thus, the results of the paper allow to perform non-/cointegration testing in typical applications like EKC, MKC or exchange rate target-zone relationships.<sup>5</sup>

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tion analysis.

A basic observation that is not always sufficiently acknowledged is the fact that nonlinear transformations change the properties of integrated processes *fundamentally*, depending upon the type of transformation. It is a basic mathematical property that only linear transformations “commute” with summation. Wagner (2012) or Stypka and Wagner (2019) exemplify this issue for polynomial transformations.

<sup>5</sup> For further discussion concerning full design and FM-OLS type estimation for cointegrating multivariate polynomial regressions (CMPRs), that potentially include arbitrary cross-products of powers of integrated

The tests discussed are illustrated with annual carbon dioxide (CO<sub>2</sub>) and sulfur dioxide (SO<sub>2</sub>) emissions data for 18 early industrialized countries over the period 1870–2016, with New Zealand data starting only in 1878. For CO<sub>2</sub> emissions an EKC relationship is found for six of the 18 countries, Austria, Belgium, Finland, Germany, Switzerland and the UK. For all countries but Germany a quadratic specification suffices, whereas for Germany a cubic specification is required. For SO<sub>2</sub> emissions an EKC relationship is found for none of the countries. In this respect note that the logarithm of GDP per capita and its powers are the only regressors included in the *simple* “*reduced-form*” equation and the absence of an EKC for SO<sub>2</sub> emissions could partly be driven by important explanatory variables and legislative changes not entirely captured by GDP. A more structural investigation of the EKC relationship is beyond the scope of this paper that is merely devoted to discussing the (non-)cointegration tests rather than to a fully fledged empirical analysis. A second observation that emerges from the illustration is that the usage of appropriate (non-)cointegration tests, designed for testing in a CPR context, reduces the evidence for a cointegrating EKC relationship compared to using (non-)cointegration tests designed for linear cointegrating relationships. By means of robustness checks we also illustrate that the results obtained with the two nonparametric tests are quite insensitive to kernel and bandwidth choices, in particular with respect to the evidence for the prevalence of an EKC. Changing the sample size and, in particular, using different data vintages lead to more variation in the test results.

The paper is organized as follows: The following section discusses and presents the tests, Sect. 3 briefly illustrates the tests with CO<sub>2</sub> and SO<sub>2</sub> emissions data, and Sect. 4 briefly summarizes and concludes. Two appendices follow the main text: “Appendix A” contains the proofs, and “Appendix B” contains the tables with the critical values. “Supplementary Appendix C” contains the detailed test results for the robustness checks.

## 2 The tests

Following Wagner and Hong (2016), we consider a cointegrating polynomial regression (CPR), i.e., an equation including a constant and polynomial time trends up to power  $q$ , integer powers of integrated regressors  $x_{jt}$ ,  $j = 1, \dots, m$  up to degrees  $p_j$  and a stationary error term  $u_t$ :

$$y_t = D_t' \theta_D + \sum_{j=1}^m X_{jt}' \theta_{X_j} + u_t, \quad \text{for } t = 1, \dots, T, \quad (1)$$

with  $D_t := [1, t, t^2, \dots, t^q]'$ ,  $X_{jt} := [x_{jt}, x_{jt}^2, \dots, x_{jt}^{p_j}]'$  and the parameter vectors  $\theta_D \in \mathbb{R}^{q+1}$  and  $\theta_{X_j} \in \mathbb{R}^{p_j}$ .

The following assumption on the regressors and the errors is put in place:

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processes as explanatory variables, see Stypka and Wagner (2020). Vogelsang and Wagner (2016) effectively consider IM-OLS estimation for CMPRs.

**Assumption 1** The process  $\{\eta_t\}_{t \in \mathbb{Z}} := \{[u_t, v_t']'\}_{t \in \mathbb{Z}}$  with  $\{v_t\}_{t \in \mathbb{Z}} := \{\Delta x_t\}_{t \in \mathbb{Z}}$ , with  $x_t := [x_{1t}, \dots, x_{mt}]'$ , is generated as:

$$\eta_t := C(L)\eta_t^0 = \sum_{j=0}^{\infty} C_j \eta_{t-j}^0,$$

with  $\sum_{j=0}^{\infty} j \|C_j\| < \infty$  and  $\det(C(1)) \neq 0$ . Furthermore, we assume that the process  $\{\eta_t^0\}_{t \in \mathbb{Z}}$  is a stationary and ergodic martingale difference sequence with natural filtration  $\mathcal{F}_t = \sigma(\{\eta_s^0\}_{s=-\infty}^t)$ , positive definite covariance matrix  $\Sigma_{\eta^0 \eta^0}$  and  $\sup_{t \geq 1} \mathbb{E}(\|\eta_t^0\|^r | \mathcal{F}_{t-1}) < \infty$  a.s. for some  $r > 4$ .

Assumption 1 is analogous to the corresponding assumption in Wagner and Hong (2016). Quite similar assumptions have been used in several places in the literature, e.g., Chang et al. (2001), Park and Phillips (1999, 2001) and Hong and Phillips (2010).<sup>6</sup> The key result that this type of assumption is required for is an invariance principle for terms of the form  $T^{-\frac{k+1}{2}} \sum_{t=1}^T x_{jt}^k u_t$ . Alternative sets of assumptions that could be used instead are formulated in Ibragimov and Phillips (2008) in a martingale framework or in de Jong (2002), Assumptions 1 and 2) who combines near epoch dependence and appropriate moment assumptions. For the present paper any such set of assumptions leading to the required invariance principle can be put in place. The assumption  $\det(C(1)) \neq 0$  together with positive definiteness of  $\Sigma_{\eta^0 \eta^0}$  implies that  $\{x_t\}$  is an integrated but not cointegrated process. Assumption 1 implies an invariance principle for  $\{\eta_t\}_{t \in \mathbb{Z}}$ :

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Tr \rfloor} \eta_t \Rightarrow B(r) = \begin{bmatrix} B_u(r) \\ B_v(r) \end{bmatrix}, \quad (2)$$

with  $B_v(r) := [B_{v_1}(r), \dots, B_{v_m}(r)]'$ . It holds that  $B(r) = \Omega^{1/2} W(r)$  with the long-run covariance matrix  $\Omega := \sum_{h=-\infty}^{\infty} \mathbb{E}(\eta_0^0 \eta_h^{0'})$  and where  $W(r)$  is an  $m+1$ -dimensional vector of standard Brownian motions. For later usage define the one-sided long-run covariance  $\Delta := \sum_{h=0}^{\infty} \mathbb{E}(\eta_0^0 \eta_h^{0'})$  and partition both matrices according to the partitioning of  $\eta_t$ :

$$\Omega = \begin{bmatrix} \Omega_{uu} & \Omega_{uv} \\ \Omega_{vu} & \Omega_{vv} \end{bmatrix}, \quad \Delta = \begin{bmatrix} \Delta_{uu} & \Delta_{uv} \\ \Delta_{vu} & \Delta_{vv} \end{bmatrix}.$$

Similarly to the linear case (compare Phillips and Hansen 1990), the OLS estimator of the parameter vector in (1) is consistent but its limiting distribution is contaminated by second-order bias terms, the presence of which renders standard asymptotic inference based on the OLS estimator invalid. Consequently, Wagner and Hong (2016, Proposition 1) extend the FM-OLS estimation principle from the cointegrating linear to the cointegrating polynomial case. The FM-OLS estimator is based

<sup>6</sup> The key difference between the assumption used here—and in Wagner and Hong (2016)—and those in the other papers mentioned is that the other papers assume that the errors  $\{u_t\}$  are serially uncorrelated, as mentioned already in Footnote 4.

on two modifications of the OLS estimator. First, the dependent variable  $y_t$  is replaced by  $y_t^+ := y_t - \Delta x_t' \hat{\Omega}_{vv}^{-1} \hat{\Omega}_{vu}$ ,  $y^+ := [y_1^+, \dots, y_T^+]'$  and, second, a correction term is subtracted. The modified dependent variable  $y_t^+$  is exactly as in Phillips and Hansen (1990), but the correction terms are different in the CPR case and given by:

$$M^* := \begin{bmatrix} M_1^* \\ \vdots \\ M_m^* \end{bmatrix}, \quad M_j^* := \hat{\Delta}_{vju}^+ \begin{bmatrix} T \\ 2 \sum x_{jt} \\ \vdots \\ p_j \sum x_{jt}^{p_j-1} \end{bmatrix}. \quad (3)$$

Throughout, we rely upon consistent estimators of the required long-run variances  $\hat{\Omega}_{vv}$ ,  $\hat{\Omega}_{vu}$ ,  $\hat{\Delta}_{vju}$  and  $\hat{\Delta}_{vju}^+ := \hat{\Delta}_{vju} - \hat{\Omega}_{uv} \hat{\Omega}_{vv}^{-1} \hat{\Delta}_{vvj}$ . In this respect, OLS consistency is important since it allows for consistent long-run variance estimation based on the OLS residuals (it is straightforward to verify that the necessary assumptions of Jansson 2002, Corollary 3 are fulfilled).

To (re-)state the asymptotic distribution of the FM-OLS estimator developed in Wagner and Hong 2016, Proposition 1), we need to define a few more quantities:  $D(r) := [1, \dots, r^q]'$ ,  $\mathbf{B}_{vj}(r) := [B_{vj}(r), \dots, B_{vj}^{p_j}(r)]'$ ,  $\mathbf{B}_v(r) := [\mathbf{B}_{v1}(r)', \dots, \mathbf{B}_{vm}(r)']'$ ,  $D := [D_1, \dots, D_T]'$ ,  $X_t := [X_{1t}', \dots, X_{mt}']'$ ,  $X := [X_1, \dots, X_T]'$ ,  $Z := [D, X]$ ,  $G_D := \text{diag}(T^{-1/2}, \dots, T^{-(q+1/2)})$ ,  $G_{X_j} := \text{diag}(T^{-1}, \dots, T^{-\frac{p_j+1}{2}})$ ,  $G_{X_m} := \text{diag}(G_{X_1}, \dots, G_{X_m})$ ,  $G := \text{diag}(G_D, G_X)$ , and  $\theta := [\theta_D', \theta_X']'$  with  $\theta_X := [\theta_{X_1}', \dots, \theta_{X_m}']'$ .

**Proposition 1** *Let  $\{y_t\}$  be generated by (1) with the regressors  $\{x_t\}$  and errors  $\{u_t\}$  satisfying Assumption 1. Define the FM-OLS estimator of  $\theta$  as:*

$$\hat{\theta}^+ := (Z'Z)^{-1} (Z'y^+ - A^*),$$

with:

$$A^* := \begin{bmatrix} 0_{(q+1) \times 1} \\ M^* \end{bmatrix}$$

and  $M^*$  as given in (3) with consistent estimators of the required long-run (co)variances. Then,  $\hat{\theta}^+$  is consistent and has a zero mean Gaussian mixture asymptotic distribution given by:

$$G^{-1} (\hat{\theta}^+ - \theta) \Rightarrow \left( \int_0^1 J(r) J(r)' dr \right)^{-1} \int_0^1 J(r) dB_{u \cdot v}(r), \quad (4)$$

with  $J(r) := [D(r)', \mathbf{B}_v(r)']'$  and  $B_{u \cdot v}(r) := B_u(r) - B_v(r)' \Omega_{vv}^{-1} \Omega_{vu}$ .

The limiting distribution given in (4) provides the basis for asymptotic standard, i.e., standard normal or chi-squared, inference on the coefficients in case the error term  $\{u_t\}$  is stationary, i.e., in case of prevalence of a CPR relationship. Thus, testing for the

presence of a CPR relationship is of prime importance and as always in cointegration testing two null hypotheses are conceivable: the null hypothesis of cointegration, i.e., the null hypothesis of stationarity of  $\{u_t\}$ , or the null hypothesis of no cointegration, i.e., testing whether (1) is in fact a spurious regression. The first null hypothesis can be tested by an extension of the Shin (1994) test, which itself is an extension of the stationarity test of Kwiatkowski et al. (1992), from cointegrating linear to cointegrating polynomial regressions. Since, of course, the errors  $u_t$  are not observed, the test statistic has to be based on observable residuals, in particular the FM-OLS residuals,  $\hat{u}_t^+$  say, can be used. The KPSS-type test statistic for the null hypothesis of cointegration is defined as:

$$CT := \frac{1}{T\hat{\Omega}_{u \cdot v}} \sum_{t=1}^T \left( \frac{1}{\sqrt{T}} \sum_{j=1}^t \hat{u}_j^+ \right)^2, \quad (5)$$

with  $\hat{\Omega}_{u \cdot v}$  a consistent estimator of the long-run variance  $\Omega_{u \cdot v} := \Omega_{uu} - \Omega_{uv}\Omega_{vv}^{-1}\Omega_{vu}$  of  $\{\hat{u}_t^+\}$ .

For testing the null hypothesis of no cointegration, we consider an extension of a variance ratio test statistic of Phillips and Ouliaris (1990), their  $\hat{P}_u$  statistic, and we continue to use this name for the extension. Denoting the OLS residuals of (1) as  $\hat{u}_t$ , the denominator of the test statistic is the properly scaled estimated “variance” of  $\{\hat{u}_t\}$ , which is an integrated process under the null hypothesis. Under the null hypothesis of a spurious regression, with  $\{y_t\}$  being an I(1) process, the stacked vector  $\{m_t\} := \{[y_t, x_t']'\}$  is a non-cointegrated I(1) vector and for later use we denote its first difference as  $\{\Delta m_t\} := \{[w_t, v_t']'\}$ . The second element of  $\hat{P}_u$  is a long-run variance estimate based on the OLS residuals of a vector autoregression (VAR) of order one for  $\{m_t\}$ , taking into account the deterministic components  $D_t$  considered:

$$m_t = \Pi_0 D_t + \Pi_1 m_{t-1} + \xi_t. \quad (6)$$

The required conditional long-run variance is computed using VAR(1) OLS residuals from (6) rather than from the vector of first differences  $\Delta m_t$ . Both versions lead to the same asymptotic behavior of the test statistic under the null hypothesis, but the asymptotic behavior differs under the alternative, compare Phillips and Ouliaris (1990, Theorems 5.2 and 5.3). Denoting the long-run variance estimated from  $\{\hat{\xi}_t\}$  as  $\tilde{\Omega} = \begin{bmatrix} \tilde{\Omega}_{ww} & \tilde{\Omega}_{wv} \\ \tilde{\Omega}_{vw} & \tilde{\Omega}_{vv} \end{bmatrix}$ , the estimated conditional long-run variance corresponding to the first component (the conditional long-run variance of  $\{w_t\}$  given  $\{v_t\}$ ) is given by  $\tilde{\Omega}_{w \cdot v} := \tilde{\Omega}_{ww} - \tilde{\Omega}_{wv}\tilde{\Omega}_{vv}^{-1}\tilde{\Omega}_{vw}$ .

The test statistic for the null hypothesis of no cointegration is defined as:

$$\hat{P}_u := \frac{\tilde{\Omega}_{w \cdot v}}{\frac{1}{T^2} \sum_{t=1}^T \hat{u}_t^2}. \quad (7)$$

The following proposition characterizes the asymptotic behavior of the two test statistics, with the *CT* Test already discussed in Wagner and Hong (2016, Proposition 5).

**Proposition 2** (i) Let  $\{y_t\}$  be generated by (1) with the regressors  $\{x_t\}$  and errors  $\{u_t\}$  satisfying Assumption 1 and let  $\hat{\Omega}_{u \cdot v}$  be a consistent estimator of  $\Omega_{u \cdot v}$ , then the asymptotic distribution of the test statistic (5) defined above is:

$$CT \Rightarrow \int_0^1 (W_{u \cdot v}^J(r))^2 dr, \quad (8)$$

with:

$$W_{u \cdot v}^J(r) := W_{u \cdot v}(r) - \int_0^r J(s)' ds \left( \int_0^1 J(s) J(s)' ds \right)^{-1} \int_0^1 J(s) dW_{u \cdot v}(s). \quad (9)$$

(ii) Let (1) be a spurious regression, i.e., let  $\{y_t\}$  be an integrated process not related to  $\{X_t\}$  in a cointegrating polynomial relationship, then the asymptotic distribution of the test statistic (7) defined above is:

$$\hat{P}_u \Rightarrow \frac{\Omega_{w \cdot v}}{\tau' \int_0^1 J^*(r) J^*(r)' dr \tau}, \quad (10)$$

with  $\tau := \begin{pmatrix} 1 \\ -(\int_0^1 J(r) J(r)' dr)^{-1} \int_0^1 J(r) B_w(r) dr \end{pmatrix}$ ,  $J^*(r) := [B_w(r), J(r)']'$  and  $\tilde{\Omega}_{w \cdot v}$  as defined above.

The asymptotic distributions of the *CT* and  $\hat{P}_u$  test statistics cannot—in general—be tabulated due to their nuisance parameter dependency related to the correlation structure between the variables. Wagner and Hong (2016, Proposition 6), following Choi and Saikkonen (2010), consider a sub-sample version of the *CT* test statistic that has a nuisance-parameter-free limiting distribution. They show, in particular, that the limiting distribution of the sub-sample *CT* test statistics is  $\int_0^1 W_{u \cdot v}(r)^2 dr$ , if the sub-samples of size  $b$  are chosen such that  $b \rightarrow \infty$  and  $b/T \rightarrow 0$ , i.e., sub-sampling achieves that the second term in (9) vanishes asymptotically. Wagner and Hong (2016), in addition, discuss how the set of sub-sample test statistics can be used in conjunction with modified Bonferroni bounds to improve the performance of the sub-sample-based test. An inspection of the proof of Proposition 1 reveals that sub-sampling does not lead to similar simplifications for the  $\hat{P}_u$  test, with this “unfortunate difference” caused by the too slow convergence rate of  $\hat{\tau}$  in the spurious regression case. Thus, for the general case only the *CT* test, when used in a sub-sampling fashion, is available.<sup>7</sup>

The problem with the nuisance parameter dependency of the limiting distributions of the test statistics originates in  $J(r)$  as defined in Proposition 1. In case it is possible

<sup>7</sup> An alternative route may be to develop bootstrap inference.



to write:

$$J(r) = \begin{bmatrix} D(r) \\ \mathbf{B}_v(r) \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & \Pi \end{bmatrix} \begin{bmatrix} D(r) \\ \mathbf{W}_v(r) \end{bmatrix}, \quad (11)$$

with a regular matrix  $\Pi$  and  $\mathbf{W}_v(r)$  a functional of standard Brownian motions one can rewrite both test statistics as functions of standard Brownian motions and a scalar (long-run) variance that can be scaled out. If a transformation as just described exists, the CPR is said to exhibit full design, a terminology coined by Vogelsang and Wagner (2016). CPRs with only one of the integrated processes present as regressor also with higher powers are straightforwardly seen to be of full design, other cases include, e.g., Translog-type relationships (see the discussion in Stypka and Wagner (Wagner et al. 2020)). For brevity, we provide the corresponding result below for the *one integrated regressor with higher power only case*, which—as mentioned before—allows to test in the empirically most relevant case covering EKC, MKC, intensity-of-use and exchange rate target-zone relationships. As before in Proposition 2, the result for the CT test is has already been discussed in Wagner and Hong (2016).

**Proposition 3** *Consider the special case of (1) when  $p_1 = \dots = p_{m-1} = 1$  and let otherwise the assumptions of Proposition 2 be fulfilled.*

- (i) *Under the null hypothesis of cointegration in (1) the limiting distribution of the CT statistic is given by:*

$$CT \Rightarrow \int_0^1 (W_{u,v}^{J^W}(r))^2 dr, \quad (12)$$

with:

$$W_{u,v}^{J^W}(r) := W_{u,v}(r) - \int_0^r J^W(s)' ds \left( \int_0^1 J^W(s) J^W(s)' ds \right)^{-1} \int_0^1 J^W(s) dW_{u,v}(s) \quad (13)$$

- and  $J^W(r) := [D(r)', \mathbf{W}(r)', W_m(r)^2, \dots, W_m(r)^{p_m}]'$ , where  $\mathbf{W}(r) := [W_1(r)', \dots, W_m(r)']'$  is a vector of standard Brownian motions independent of  $W_{u,v}(r)$ .
- (ii) *Under the null hypothesis of (1) being a spurious regression, the limiting distribution of the  $\hat{P}_u$  statistic is given by:*

$$\hat{P}_u \Rightarrow \frac{1}{\int_0^1 W_{w,v}^2(r) dr - \int_0^1 W_{w,v}(r) J^W(r)' dr \left( \int_0^1 J^W(r) J^W(r)' dr \right)^{-1} \int_0^1 J^W(r) W_{w,v}(r) dr}, \quad (14)$$

with  $W_{w,v}(r)$  independent of  $\mathbf{W}(r)$ .

The limiting distributions (12) and (14) can be tabulated because they are functionals of standard Brownian motions. The corresponding critical values depend upon the deterministic component, the number of integrated regressors and the included powers

**Table 1** List of countries included in the empirical analysis

|           |         |         |             |             |         |
|-----------|---------|---------|-------------|-------------|---------|
| Australia | Austria | Belgium | Canada      | Denmark     | Finland |
| France    | Germany | Italy   | Japan       | New Zealand | Norway  |
| Portugal  | Spain   | Sweden  | Switzerland | UK          | USA     |

The sample range is 1870–2016 with the exception of New Zealand with sample range 1878–2016

of the single integrated regressor that enters the CPR with higher powers. Critical values for the  $CT$  statistic are given in Table 6 and for the  $\hat{P}_u$  statistic in Table 7 in “Appendix B.” These tables extend the corresponding tables provided in Shin (1994) and Phillips and Ouliaris (1990) for linear cointegrating relationships to the CPR case. The tables contain critical values for up to four integrated regressors, up to power four of the integrated regressor entering with powers for the usual deterministic components, i.e., no deterministic component, an intercept only and intercept and linear trend.<sup>8</sup>

### 3 An illustration with the environmental Kuznets curve

We illustrate the tests discussed with annual data for real GDP per capita, carbon dioxide ( $\text{CO}_2$ ) in metric tons per capita and sulfur dioxide ( $\text{SO}_2$ ) emissions in kilograms per capita for 18 early industrialized countries, listed in Table 1, over the period 1870–2016; with the exception of New Zealand where the sample range is 1878–2016. Real GDP is measured in 2011 US dollars and calculated from the GDP and population data in the 2018 version of the Maddison Project Database (see Bolt et al. 2018). The  $\text{CO}_2$  per capita emissions are calculated from total  $\text{CO}_2$  emissions from fossil fuel usage, downloaded from the webpage of the Carbon Dioxide Information Analysis Center (CDIAC) at the Appalachian State University (see Boden et al. 2018). The  $\text{SO}_2$  per capita emissions are calculated from combining total  $\text{SO}_2$  emissions taken from the NASA Socioeconomic Data and Applications Center (see Smith et al. 2011), which provides data for 1870–2005, with Emissions (2020) data for the period 2006–2016.<sup>9</sup>

We use the data described in logarithmic specifications and test for cointegration and non-cointegration in both the quadratic and the cubic EKC relationship including an intercept and a linear trend:

$$e_t = c + \delta t + \beta_1 y_t + \beta_2 y_t^2 + u_t, \quad (15)$$

$$e_t = c + \delta t + \beta_1 y_t + \beta_2 y_t^2 + \beta_3 y_t^3 + u_t, \quad (16)$$

with  $e_t$  denoting the logarithm of emissions ( $\text{CO}_2$  or  $\text{SO}_2$ ) per capita and  $y_t$  the logarithm of GDP per capita.

<sup>8</sup> MATLAB code to generate (additional) critical values as well as to perform the tests is available upon request.

<sup>9</sup> To be precise, the OECD provides data for  $\text{SO}_x$  emissions. The National (1975) estimates the share of  $\text{SO}_2$  in  $\text{SO}_x$  emissions at about 98% and this is the factor we also use to transform OECD  $\text{SO}_x$  to  $\text{SO}_2$  data.

Performing the usual battery of unit root tests on the logarithm of GDP per capita does not lead to a rejection of the null hypothesis for any of the 18 countries. Thus, a “necessary condition” for embarking on CPR analysis, that of an integrated regressor, appears to be fulfilled. The *CT* test is performed using the FM-OLS residuals (i.e., is based on using the estimator of Wagner and Hong 2016) with the Bartlett kernel and the Newey and West (1994) bandwidth rule.<sup>10</sup>

Tables 2 and 3 contain the test results for CO<sub>2</sub> and SO<sub>2</sub> emissions, respectively. We use the following (quite obvious) “decision rule”: If *CT* does not lead to a rejection, but  $\hat{P}_u$  does, this is evidence for the presence of a cointegrating polynomial EKC relationship. If *CT* does lead to a rejection, but  $\hat{P}_u$  does not, this is evidence against the presence of an EKC. If either both or none of the tests lead to a rejection, this is regarded as conflicting evidence.<sup>11</sup> Given this decision rule, let us now turn to the results, first for CO<sub>2</sub> emissions given in Table 2. For six of the 18 countries, Austria, Belgium, Finland, Germany, Switzerland, and the UK we find evidence for an EKC relationship. For Germany, only for the cubic specification, for the other five countries for both the quadratic and the nesting cubic specification (which is, of course, a necessary theoretical implication of the prevalence of a quadratic CPR). The test decisions are also consistent between the cubic and quadratic specifications in the sense that evidence against the presence of a cubic EKC for a certain country comes along with evidence also against a quadratic EKC for that country. The one exception here is Sweden for which there is evidence against an EKC for the cubic specification and “only” conflicting evidence for the quadratic specification. Also note that conflicting evidence for the cubic specification occurs typically in conjunction with evidence against a quadratic EKC. The results for SO<sub>2</sub> emissions given in Table 3 are quite clear: There is no evidence for a quadratic or cubic EKC for any of the 18 countries considered. For 16 of the 18 countries, however, the evidence is conflicting for the cubic specification.

The limited evidence for the prevalence of an EKC is akin to an observation made earlier in Wagner (2015), i.e., that the usage of adequate (non-)cointegration tests reduces the evidence for an EKC compared to using (non-)cointegration tests (critical values) developed for linear cointegrating relationships. This is exemplified here by including also the *asymptotically invalid* test results obtained by applying the residual-based augmented Dickey–Fuller non-cointegration *t*-type test of Phillips and Ouliaris (1990) developed for cointegrating linear relationships, labeled  $PO_t$ . Using this test, for CO<sub>2</sub> emissions the null hypothesis of non-cointegration is rejected for twelve of the 18 countries at the 5% significance level compared to only five rejections obtained with the asymptotically valid  $\hat{P}_u$  test. For SO<sub>2</sub> emissions, the differences between the  $\hat{P}_u$  and  $PO_t$  results are less pronounced, but point in the same direction. These findings illustrate the importance of using (non-)cointegration tests—or more precisely

<sup>10</sup> The following subsection investigates inter alia the impact of kernel and bandwidth choices on the test results.

<sup>11</sup> In those cases where one of the tests rejects at the 5% significance level, but the other only at the 10% significance level, we put higher weight on the “stronger” 5% rejection. Considering also such cases as conflicting evidence does not lead to substantial differences in the overall picture. In Tables 4 and 5 that summarize the results from the robustness checks, evidence for an EKC is labeled with “y,” evidence against an EKC with “n” and conflicting evidence with “o.”

**Table 2** Cointegration ( $CT$ ) and non-cointegration ( $\hat{P}_u$ ) test results for CO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |               |                | Cubic        |                |                |
|-------------|--------------|---------------|----------------|--------------|----------------|----------------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$         | $CT$         | $\hat{P}_u$    | $PO_t$         |
| Australia   | <b>0.107</b> | 11.294        | −2.623         | <b>0.106</b> | 11.306         | −2.530         |
| Austria     | 0.055        | <b>57.186</b> | −3.816         | 0.042        | <b>57.988</b>  | −3.822         |
| Belgium     | 0.061        | <i>51.074</i> | − <b>5.710</b> | 0.057        | <i>55.199</i>  | − <b>5.667</b> |
| Canada      | <b>0.144</b> | 11.882        | −3.355         | 0.056        | 25.542         | − <b>4.878</b> |
| Denmark     | 0.047        | 36.027        | − <b>4.673</b> | 0.049        | 36.029         | − <b>4.670</b> |
| Finland     | 0.048        | <b>75.487</b> | − <b>5.719</b> | 0.033        | <b>83.542</b>  | − <b>6.136</b> |
| France      | 0.066        | 27.859        | − <b>4.929</b> | 0.062        | 28.041         | − <b>4.863</b> |
| Germany     | <b>0.108</b> | <b>69.262</b> | − <b>8.001</b> | <i>0.090</i> | <b>69.349</b>  | − <b>8.089</b> |
| Italy       | <b>0.138</b> | 34.120        | − <b>4.174</b> | <i>0.088</i> | <i>50.661</i>  | − <b>5.520</b> |
| Japan       | <b>0.149</b> | 8.718         | − <b>5.889</b> | 0.065        | 12.995         | − <b>6.009</b> |
| New Zealand | <b>0.112</b> | 13.698        | − <b>5.422</b> | <i>0.098</i> | 14.203         | − <b>5.809</b> |
| Norway      | <b>0.116</b> | 20.583        | −3.334         | <i>0.092</i> | 26.063         | −3.617         |
| Portugal    | <b>0.108</b> | 20.919        | − <b>9.181</b> | <b>0.111</b> | 21.327         | − <b>9.412</b> |
| Spain       | <i>0.091</i> | 42.031        | −3.315         | <i>0.089</i> | 42.112         | −3.384         |
| Sweden      | 0.084        | 29.150        | − <b>4.312</b> | <i>0.084</i> | 29.988         | − <i>4.323</i> |
| Switzerland | <i>0.097</i> | <b>86.476</b> | − <b>6.291</b> | 0.056        | <b>107.442</b> | − <b>6.763</b> |
| UK          | 0.070        | <b>91.459</b> | − <b>6.894</b> | 0.067        | <b>91.704</b>  | − <b>6.848</b> |
| USA         | <b>0.148</b> | 12.620        | −2.337         | 0.077        | 23.531         | −3.558         |

The columns  $PO_t$  display the test results of the residual-based augmented Dickey–Fuller non-cointegration  $t$ -type test of Phillips and Ouliaris (1990) developed for cointegrating linear relationships. *Italic* entries indicate rejection of the null hypothesis at the 10% level and bold entries rejection at the 5% level. The left panel displays the results for the quadratic specification and the right panel for the cubic specification. For both polynomial degrees intercept and linear trend are included. The results are based on the Bartlett kernel with bandwidth chosen to Newey and West (1994)

appropriate critical values—constructed for the CPR setting. For the  $CT$  test, this issue has already been illustrated in Wagner (2015) and is discussed more thoroughly from a theoretical perspective in Stypka et al. (2019).

### 3.1 Robustness checks

Both tests, being nonparametric, depend upon kernel and bandwidth choices. Consequently, the results might be sensitive to these choices. To gauge the extent of this potential sensitivity, we consider four combinations of bandwidths and kernels. The Bartlett kernel and the Newey and West (1994) bandwidth (the baseline, used above), labeled I, the Bartlett kernel and the Andrews (1991) bandwidth, labeled II, and the Quadratic Spectral kernel with these two bandwidth rules, labeled III and IV in Tables 4 for CO<sub>2</sub> emissions and 5 for SO<sub>2</sub> emissions. It turns out that the results are in fact highly robust with respect to these choices, most importantly the evidence for the prevalence of an EKC for CO<sub>2</sub> emissions. For the five countries with evidence for a

**Table 3** Cointegration ( $CT$ ) and non-cointegration  $\hat{P}_u$  test results for SO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |             |               | Cubic        |             |               |
|-------------|--------------|-------------|---------------|--------------|-------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$        | $CT$         | $\hat{P}_u$ | $PO_t$        |
| Australia   | 0.051        | 38.885      | −3.402        | 0.048        | 40.506      | −3.521        |
| Austria     | 0.080        | 28.695      | −3.776        | 0.055        | 39.188      | −3.695        |
| Belgium     | 0.100        | 17.752      | −3.738        | 0.043        | 27.436      | −3.556        |
| Canada      | 0.055        | 22.976      | −3.865        | 0.053        | 23.118      | −3.888        |
| Denmark     | 0.101        | 10.010      | −3.160        | 0.057        | 20.890      | −4.006        |
| Finland     | 0.078        | 25.292      | −4.082        | 0.077        | 25.701      | −3.905        |
| France      | <b>0.112</b> | 5.524       | −2.353        | 0.067        | 11.725      | −2.455        |
| Germany     | 0.098        | 6.057       | −2.584        | 0.062        | 11.524      | −2.782        |
| Italy       | 0.089        | 3.835       | −2.296        | 0.090        | 4.090       | −1.729        |
| Japan       | <b>0.113</b> | 15.237      | −3.219        | 0.036        | 19.190      | −4.447        |
| New Zealand | 0.090        | 11.831      | <b>−4.849</b> | 0.054        | 16.130      | <b>−6.941</b> |
| Norway      | 0.069        | 22.719      | −3.216        | 0.067        | 22.931      | −3.283        |
| Portugal    | 0.100        | 11.880      | −3.175        | <b>0.102</b> | 11.933      | −3.271        |
| Spain       | 0.074        | 12.060      | −1.799        | 0.080        | 13.046      | −1.761        |
| Sweden      | 0.066        | 11.155      | −3.709        | 0.067        | 12.278      | −2.908        |
| Switzerland | <b>0.203</b> | 6.766       | −1.703        | 0.060        | 37.684      | <b>−5.067</b> |
| UK          | 0.075        | 0.841       | −1.510        | 0.066        | 1.731       | −2.647        |
| USA         | 0.085        | 10.517      | −1.230        | 0.074        | 10.574      | −1.200        |

For further explanations see caption of Table 2

quadratic (and cubic) EKC, this evidence is present for all four combinations of kernel and bandwidth. For Germany, with evidence for the cubic specification only and mixed evidence for the quadratic specification in the baseline combination I, the evidence is scattered a bit between “y” and “o” across kernel and bandwidth choices. Looking at columns I to IV in Table 4 indicates some limited variation between “n” and “o” across combinations for some of the countries. For SO<sub>2</sub> emissions, there is—similarly to CO<sub>2</sub> emissions—some variability for some countries between “n” and “o” decisions. However, for Switzerland using the Andrews (1991) bandwidth rule leads to evidence for an EKC in the cubic specification, whereas the evidence is conflicting when using the Newey and West (1994) bandwidth rule. The differences in conclusions are driven by  $\hat{P}_u$  rejecting with the Andrews (1991) bandwidth. Altogether, however, the test results are quite insensitive—most importantly with respect to evidence for the prevalence of an EKC—to kernel and bandwidth choices.

Columns N<sub>0</sub> and O<sub>0</sub> of Tables 4 and 5 shed light on two other *empirical* dimensions of robustness of the test results: N<sub>0</sub> with respect to the sample size, using the data used so far but only until 2000 and O<sub>0</sub> using the same data as used in Wagner (2015). Sample size and data vintage robustness do not directly relate to robustness of the tests, but it may still be informative to see their impact on the test results for our EKC illustration. The results in columns N<sub>0</sub> and O<sub>0</sub> are based on using the Bartlett kernel and the Newey

**Table 4** Test decisions, as explained above, for CO<sub>2</sub> emissions, with “y” indicating evidence for the presence of an EKC, “n” evidence against and “o” indicating conflicting evidence

|             | Quadratic |    |     |    |                |                | Cubic |    |     |    |                |                |
|-------------|-----------|----|-----|----|----------------|----------------|-------|----|-----|----|----------------|----------------|
|             | I         | II | III | IV | N <sub>0</sub> | O <sub>0</sub> | I     | II | III | IV | N <sub>0</sub> | O <sub>0</sub> |
| Australia   | n         | o  | n   | o  | n              | n              | n     | o  | n   | n  | n              | n              |
| Austria     | y         | y  | y   | y  | y              | y              | y     | y  | y   | y  | y              | y              |
| Belgium     | y         | y  | o   | y  | y              | y              | y     | y  | o   | y  | y              | y              |
| Canada      | n         | n  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |
| Denmark     | o         | o  | o   | o  | o              | o              | o     | o  | o   | o  | o              | o              |
| Finland     | y         | y  | y   | y  | y              | y              | y     | y  | y   | y  | y              | y              |
| France      | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| Germany     | o         | o  | y   | o  | o              | o              | y     | o  | y   | y  | y              | o              |
| Italy       | n         | n  | n   | o  | n              | n              | o     | o  | n   | o  | o              | n              |
| Japan       | n         | n  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |
| New Zealand | n         | o  | n   | o  | n              | n              | n     | o  | n   | o  | n              | n              |
| Norway      | n         | o  | n   | o  | n              | n              | n     | o  | n   | o  | n              | n              |
| Portugal    | n         | n  | n   | n  | n              | n              | n     | n  | n   | n  | n              | n              |
| Spain       | n         | o  | n   | o  | n              | n              | n     | o  | n   | o  | n              | n              |
| Sweden      | o         | o  | o   | o  | o              | n              | n     | o  | n   | o  | n              | n              |
| Switzerland | y         | y  | y   | y  | y              | n              | y     | y  | y   | y  | y              | n              |
| UK          | y         | y  | y   | y  | y              | o              | y     | y  | y   | y  | y              | o              |
| USA         | n         | o  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |

I, II, III and IV indicate test results based on different combinations of kernels and bandwidths as detailed in the main text. N<sub>0</sub> indicates results based on the data used in this paper until 2000 only and O<sub>0</sub> indicates test results using an earlier data vintage until 2000. All test results using data until 2000 are based on the Bartlett kernel and the Newey and West (1994) bandwidth rule

and West (1994) bandwidth. Comparing columns N<sub>0</sub> with columns I leads to only one difference, for SO<sub>2</sub> emissions of Canada in the quadratic specification. Using the full data range the evidence points against the prevalence of an EKC, whereas the reduced sample leads to conflicting evidence. Thus, it appears that the additional 16 years of data help to sharpen inference. We close the robustness check by comparing the results for the sample period until 2000 with the results obtained using the data of Wagner (2015) that also range until 2000, i.e., we compare the results in columns N<sub>0</sub> and O<sub>0</sub>. For CO<sub>2</sub> emissions, using the new data vintage leads to evidence for the prevalence of an EKC for three more countries than for the old data vintage, for Switzerland and the UK for both specifications and for Germany for the cubic specification. Additionally, the evidence for four countries is changed from negative to conflicting, with these changes occurring primarily in the cubic specification. For SO<sub>2</sub> emissions the evidence changes from negative to conflicting for a number of countries in both the quadratic and the cubic specification. Additionally, the evidence changes for the UK from positive to conflicting for the cubic specification.<sup>12</sup> Whether these changes relate to some form

<sup>12</sup> The UK was the sole country for which Wagner (2015) finds an EKC for SO<sub>2</sub> emissions.

**Table 5** Test decisions for SO<sub>2</sub> emissions, emissions, with “y” indicating evidence for the presence of an EKC, “n” evidence against and “o” indicating conflicting evidence

|             | Quadratic |    |     |    |                |                | Cubic |    |     |    |                |                |
|-------------|-----------|----|-----|----|----------------|----------------|-------|----|-----|----|----------------|----------------|
|             | I         | II | III | IV | N <sub>0</sub> | O <sub>0</sub> | I     | II | III | IV | N <sub>0</sub> | O <sub>0</sub> |
| Australia   | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| Austria     | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| Belgium     | n         | o  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |
| Canada      | n         | o  | o   | o  | o              | o              | o     | o  | o   | o  | o              | o              |
| Denmark     | n         | o  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |
| Finland     | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| France      | n         | n  | n   | n  | n              | n              | o     | n  | o   | n  | o              | n              |
| Germany     | n         | o  | n   | n  | n              | n              | o     | o  | o   | n  | o              | n              |
| Italy       | n         | n  | o   | n  | n              | n              | n     | n  | n   | n  | n              | n              |
| Japan       | n         | o  | n   | o  | n              | n              | o     | o  | o   | o  | o              | n              |
| New Zealand | n         | o  | n   | n  | n              | n              | o     | o  | o   | o  | o              | n              |
| Norway      | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| Portugal    | n         | o  | n   | o  | n              | n              | n     | o  | n   | n  | n              | n              |
| Spain       | o         | n  | o   | n  | o              | n              | o     | n  | o   | n  | o              | n              |
| Sweden      | o         | o  | o   | o  | o              | n              | o     | o  | o   | o  | o              | n              |
| Switzerland | n         | n  | n   | n  | n              | n              | o     | y  | o   | y  | o              | n              |
| UK          | o         | n  | o   | n  | o              | n              | o     | o  | o   | o  | o              | y              |
| USA         | o         | n  | o   | n  | o              | n              | o     | n  | o   | n  | o              | n              |

For further explanations see caption to Table 4

of sensitivity of the tests to data with unchanged CPR characteristics or because the data revision truly changes the nature of the data is, of course, not clear. What we can, however, take home from our robustness checks is that the tests are not very sensitive to kernel and bandwidth choices.

## 4 Summary and conclusions

This paper discusses two nonparametric tests: one for the null hypothesis of a cointegrating polynomial regression and one for the null hypothesis of the absence of cointegration. Both tests are extensions from tests developed for cointegrating linear regressions to the cointegrating polynomial regression setting. Specifically, they are extensions of the Shin (1994) test and of a variance ratio test of Phillips and Ouliaris (1990).<sup>13</sup> The word extension hereby refers to deriving the null limiting distributions of the test statistics in the CPR setting. It turns out that, in the general case, neither of the two tests has a nuisance-parameter-free limiting distribution. However, in case

<sup>13</sup> Note again that the  $CT$  test has already been discussed in Wagner and Hong (2016) and used in Wagner (2015). Both, the  $CT$  and the  $\hat{P}_u$  test have already been used in an analysis of the material Kuznets Curve hypothesis in Grabarczyk et al. (2018).

of *full design* of the regression model, both tests' limiting distributions are nuisance parameter free and can hence be tabulated. The empirically most relevant case with only one of the integrated regressors appearing also with higher powers is easily seen to be of full design. For this case, "Appendix B" provides tables with critical values for up to four integrated regressors, up to power four of the single integrated regressor entering with higher powers and the usual deterministic specifications.

The tests are briefly illustrated—which is by no means a fully fledged EKC analysis—for the environmental Kuznets curve for both CO<sub>2</sub> and SO<sub>2</sub> emissions for 18 early industrialized countries over, with the exception of New Zealand, the period 1870–2016. For CO<sub>2</sub> emissions an EKC relationship appears to be present for six of the 18 countries, i.e., Austria, Belgium, Finland, Germany, Switzerland and the UK. For SO<sub>2</sub> emissions there is no evidence for the prevalence of a quadratic or cubic cointegrating EKC relationship. Little or reduced evidence for an EKC is a typical finding when using adequate tests compared to using tests designed for linear cointegrating relationships, as is discussed in this paper and already earlier in Wagner (2015). The illustration also indicates that the test results are not (really) sensitive with respect to kernel and bandwidth choices, most importantly the results for the prevalence of an EKC. Our illustration also shows that the findings are more sensitive with respect to sample size and in particular data vintage used.

The tests discussed are important inputs for developing (non-)cointegration tests in panel (see, e.g., de Jong and Wagner 2022; Wagner and Reichold 2022) or seemingly unrelated CPR settings (see, e.g., Wagner et al. 2020).

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## Declaration

**Conflict of interest** The author declares that he has no conflict of interest related to this research. This article does not contain any studies with human participants or animals performed by the author.

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## Appendix A: Proofs

**Proof of Proposition 2** The proof of item (i) is contained in Wagner and Hong (2016, Proposition 5) and it thus only remains to show the second item. By definition, it holds that:

$$\hat{u}_t = (y_t, Z'_t) \begin{pmatrix} 1 \\ -\hat{\theta} \end{pmatrix} =: \tilde{Z}'_t \hat{\tau}, \quad (17)$$

**Table 6** Critical values for the *CT* test for the case of only one regressor entering the CPR with powers

|                            | 0.010  | 0.025  | 0.050  | 0.100  | 0.500  | 0.900  | 0.950  | 0.975  | 0.990  |
|----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <i>m</i> = 1, <i>p</i> = 1 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0283 | 0.0354 | 0.0438 | 0.0576 | 0.2016 | 0.8575 | 1.2087 | 1.6005 | 2.1413 |
| c,t                        | 0.0206 | 0.0247 | 0.0292 | 0.0357 | 0.0832 | 0.2325 | 0.3164 | 0.4084 | 0.5495 |
| c,t                        | 0.0155 | 0.0180 | 0.0206 | 0.0243 | 0.0467 | 0.0977 | 0.1212 | 0.1450 | 0.1786 |
| <i>m</i> = 1, <i>p</i> = 2 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0247 | 0.0311 | 0.0379 | 0.0492 | 0.1591 | 0.6639 | 0.9465 | 1.2649 | 1.7119 |
| c                          | 0.0188 | 0.0224 | 0.0263 | 0.0321 | 0.0742 | 0.2133 | 0.2927 | 0.3786 | 0.5044 |
| c,t                        | 0.0144 | 0.0165 | 0.0189 | 0.0221 | 0.0416 | 0.0858 | 0.1063 | 0.1280 | 0.1568 |
| <i>m</i> = 1, <i>p</i> = 3 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0229 | 0.0282 | 0.0346 | 0.0442 | 0.1382 | 0.5605 | 0.8039 | 1.0792 | 1.4731 |
| c                          | 0.0177 | 0.0210 | 0.0246 | 0.0299 | 0.0699 | 0.2044 | 0.2807 | 0.3651 | 0.4904 |
| c,t                        | 0.0137 | 0.0157 | 0.0179 | 0.0209 | 0.0391 | 0.0812 | 0.1010 | 0.1221 | 0.1501 |
| <i>m</i> = 1, <i>p</i> = 4 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0216 | 0.0266 | 0.0322 | 0.0413 | 0.1262 | 0.4962 | 0.7079 | 0.9386 | 1.2753 |
| c                          | 0.0169 | 0.0200 | 0.0234 | 0.0286 | 0.0673 | 0.1987 | 0.2743 | 0.3571 | 0.4767 |
| c,t                        | 0.0131 | 0.0151 | 0.0172 | 0.0201 | 0.0376 | 0.0784 | 0.0977 | 0.1180 | 0.1455 |
| <i>m</i> = 2, <i>p</i> = 1 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0239 | 0.0296 | 0.0361 | 0.0465 | 0.1496 | 0.6257 | 0.8939 | 1.1890 | 1.6157 |
| c                          | 0.0179 | 0.0211 | 0.0247 | 0.0297 | 0.0633 | 0.1615 | 0.2194 | 0.2853 | 0.3844 |
| c,t                        | 0.0140 | 0.0162 | 0.0185 | 0.0216 | 0.0401 | 0.0819 | 0.1014 | 0.1215 | 0.1495 |
| <i>m</i> = 2, <i>p</i> = 2 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0217 | 0.0265 | 0.0318 | 0.0405 | 0.1232 | 0.5046 | 0.7253 | 0.9711 | 1.3463 |
| c                          | 0.0165 | 0.0194 | 0.0224 | 0.0269 | 0.0570 | 0.1479 | 0.2006 | 0.2631 | 0.3586 |
| c,t                        | 0.0131 | 0.0151 | 0.0171 | 0.0198 | 0.0362 | 0.0726 | 0.0900 | 0.1080 | 0.1334 |
| <i>m</i> = 2, <i>p</i> = 3 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0202 | 0.0244 | 0.0292 | 0.0370 | 0.1085 | 0.4422 | 0.6384 | 0.8546 | 1.1769 |
| c                          | 0.0155 | 0.0182 | 0.0211 | 0.0253 | 0.0535 | 0.1408 | 0.1914 | 0.2532 | 0.3484 |
| c,t                        | 0.0125 | 0.0143 | 0.0162 | 0.0188 | 0.0341 | 0.0687 | 0.0852 | 0.1029 | 0.1278 |
| <i>m</i> = 2, <i>p</i> = 4 |        |        |        |        |        |        |        |        |        |
| c                          | 0.0189 | 0.0231 | 0.0276 | 0.0349 | 0.0995 | 0.4002 | 0.5725 | 0.7652 | 1.0554 |

Table 6 continued

|                | 0.010  | 0.025  | 0.050  | 0.100  | 0.500  | 0.900  | 0.950  | 0.975  | 0.990  |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| c              | 0.0149 | 0.0174 | 0.0202 | 0.0243 | 0.0515 | 0.1366 | 0.1864 | 0.2476 | 0.3391 |
| c,t            | 0.0120 | 0.0138 | 0.0155 | 0.0181 | 0.0328 | 0.0663 | 0.0822 | 0.0997 | 0.1244 |
| $m = 3, p = 1$ |        |        |        |        |        |        |        |        |        |
|                | 0.0208 | 0.0254 | 0.0305 | 0.0386 | 0.1174 | 0.4722 | 0.6780 | 0.9003 | 1.2361 |
| c              | 0.0158 | 0.0184 | 0.0213 | 0.0252 | 0.0508 | 0.1204 | 0.1588 | 0.2062 | 0.2770 |
| c,t            | 0.0129 | 0.0147 | 0.0167 | 0.0193 | 0.0348 | 0.0689 | 0.0851 | 0.1022 | 0.1270 |
| $m = 3, p = 2$ |        |        |        |        |        |        |        |        |        |
|                | 0.0190 | 0.0230 | 0.0275 | 0.0345 | 0.0989 | 0.3920 | 0.5617 | 0.7589 | 1.0433 |
| c              | 0.0146 | 0.0170 | 0.0197 | 0.0232 | 0.0459 | 0.1095 | 0.1457 | 0.1906 | 0.2593 |
| c,t            | 0.0121 | 0.0138 | 0.0155 | 0.0179 | 0.0317 | 0.0615 | 0.0754 | 0.0905 | 0.1123 |
| $m = 3, p = 3$ |        |        |        |        |        |        |        |        |        |
|                | 0.0178 | 0.0214 | 0.0255 | 0.0317 | 0.0885 | 0.3499 | 0.5039 | 0.6809 | 0.9435 |
| c              | 0.0140 | 0.0162 | 0.0186 | 0.0220 | 0.0433 | 0.1041 | 0.1391 | 0.1819 | 0.2477 |
| c,t            | 0.0115 | 0.0132 | 0.0148 | 0.0170 | 0.0300 | 0.0583 | 0.0718 | 0.0864 | 0.1068 |
| $m = 3, p = 4$ |        |        |        |        |        |        |        |        |        |
|                | 0.0170 | 0.0204 | 0.0242 | 0.0300 | 0.0823 | 0.3207 | 0.4613 | 0.6185 | 0.8738 |
| c              | 0.0134 | 0.0156 | 0.0179 | 0.0211 | 0.0417 | 0.1012 | 0.1357 | 0.1774 | 0.2425 |
| c,t            | 0.0112 | 0.0127 | 0.0143 | 0.0164 | 0.0288 | 0.0564 | 0.0696 | 0.0838 | 0.1039 |
| $m = 4, p = 1$ |        |        |        |        |        |        |        |        |        |
|                | 0.0184 | 0.0222 | 0.0265 | 0.0333 | 0.0960 | 0.3754 | 0.5327 | 0.7093 | 0.9688 |
| c              | 0.0143 | 0.0165 | 0.0189 | 0.0222 | 0.0422 | 0.0945 | 0.1221 | 0.1537 | 0.2027 |
| c,t            | 0.0119 | 0.0136 | 0.0153 | 0.0176 | 0.0308 | 0.0591 | 0.0727 | 0.0877 | 0.1088 |
| $m = 4, p = 2$ |        |        |        |        |        |        |        |        |        |
|                | 0.0170 | 0.0204 | 0.0243 | 0.0302 | 0.0829 | 0.3173 | 0.4483 | 0.6050 | 0.8326 |
| c              | 0.0133 | 0.0155 | 0.0175 | 0.0205 | 0.0386 | 0.0858 | 0.1108 | 0.1400 | 0.1876 |
| c,t            | 0.0113 | 0.0128 | 0.0143 | 0.0164 | 0.0282 | 0.0534 | 0.0653 | 0.0785 | 0.0969 |
| $m = 4, p = 3$ |        |        |        |        |        |        |        |        |        |
|                | 0.0161 | 0.0193 | 0.0228 | 0.0280 | 0.0748 | 0.2844 | 0.4089 | 0.5494 | 0.7540 |
| c              | 0.0128 | 0.0147 | 0.0167 | 0.0195 | 0.0365 | 0.0815 | 0.1058 | 0.1341 | 0.1802 |
| c,t            | 0.0109 | 0.0122 | 0.0137 | 0.0157 | 0.0268 | 0.0508 | 0.0623 | 0.0745 | 0.0925 |
| $m = 4, p = 4$ |        |        |        |        |        |        |        |        |        |
|                | 0.0154 | 0.0183 | 0.0216 | 0.0265 | 0.0698 | 0.2636 | 0.3788 | 0.5055 | 0.6930 |
| c              | 0.0123 | 0.0142 | 0.0161 | 0.0188 | 0.0352 | 0.0787 | 0.1028 | 0.1316 | 0.1762 |
| c,t            | 0.0105 | 0.0118 | 0.0132 | 0.0151 | 0.0258 | 0.0490 | 0.0600 | 0.0718 | 0.0887 |

The symbols in the first column indicate the deterministic component: none (empty), intercept only (c) and intercept and linear trend (c,t),  $m$  indicates the number of integrated regressors, and  $p$  indicates the highest included power of the regressor entering with powers

**Table 7** Critical values for the  $\hat{P}_u$  test for the null hypothesis of no cointegration for the case of only one regressor entering the CPR with powers

|                | 0.010  | 0.025   | 0.050   | 0.100   | 0.500   | 0.900   | 0.950   | 0.975   | 0.990   |
|----------------|--------|---------|---------|---------|---------|---------|---------|---------|---------|
| $m = 1, p = 1$ |        |         |         |         |         |         |         |         |         |
|                | 0.5845 | 0.8326  | 1.1653  | 1.7665  | 7.2292  | 20.5040 | 26.1196 | 31.6526 | 39.0403 |
| c              | 1.8825 | 2.4752  | 3.1790  | 4.3116  | 11.9696 | 27.7789 | 34.0970 | 40.1474 | 48.0348 |
| c,t            | 5.5163 | 6.8023  | 8.1928  | 10.1515 | 21.4332 | 41.1315 | 48.5640 | 55.3291 | 64.4630 |
| $m = 1, p = 2$ |        |         |         |         |         |         |         |         |         |
|                | 0.8033 | 1.1261  | 1.5465  | 2.2889  | 8.9577  | 24.5044 | 30.8754 | 36.9862 | 44.8619 |
| c              | 2.0263 | 2.6460  | 3.4311  | 4.6944  | 13.4370 | 30.9825 | 37.8748 | 44.4325 | 52.9478 |
| c,t            | 6.2801 | 7.7521  | 9.3365  | 11.5826 | 24.0819 | 45.2373 | 52.9518 | 60.3206 | 69.4706 |
| $m = 1, p = 3$ |        |         |         |         |         |         |         |         |         |
|                | 0.9509 | 1.3362  | 1.8350  | 2.6904  | 10.0724 | 27.0794 | 33.8596 | 40.5755 | 48.9771 |
| c              | 2.0991 | 2.7549  | 3.5724  | 4.8907  | 14.2797 | 33.1538 | 40.4370 | 47.4776 | 56.1230 |
| c,t            | 6.5576 | 8.1182  | 9.8120  | 12.2592 | 25.5527 | 47.9252 | 55.9264 | 63.5980 | 73.3074 |
| $m = 1, p = 4$ |        |         |         |         |         |         |         |         |         |
|                | 1.0791 | 1.5079  | 2.0546  | 3.0064  | 10.8663 | 28.8801 | 36.1806 | 43.2061 | 51.8041 |
| c              | 2.1442 | 2.8181  | 3.6672  | 5.0256  | 14.8495 | 34.7177 | 42.4254 | 49.7957 | 58.7888 |
| c,t            | 6.7354 | 8.3833  | 10.1501 | 12.7102 | 26.6210 | 50.0143 | 58.2791 | 66.0470 | 76.1264 |
| $m = 2, p = 1$ |        |         |         |         |         |         |         |         |         |
|                | 1.0623 | 1.6048  | 2.2985  | 3.4434  | 11.2203 | 27.2499 | 33.5122 | 39.8156 | 47.7341 |
| c              | 2.5816 | 3.4984  | 4.5844  | 6.1575  | 15.7656 | 33.8495 | 40.9039 | 47.6771 | 55.7996 |
| c,t            | 6.5444 | 8.1968  | 9.8377  | 12.1579 | 25.0446 | 46.3967 | 54.2696 | 61.8035 | 71.4986 |
| $m = 2, p = 2$ |        |         |         |         |         |         |         |         |         |
|                | 1.3404 | 1.9743  | 2.7902  | 4.1475  | 13.2952 | 31.1699 | 38.0363 | 44.8716 | 53.2264 |
| c              | 2.7868 | 3.7925  | 5.0189  | 6.7601  | 17.6129 | 37.2768 | 44.7916 | 51.8411 | 60.6054 |
| c,t            | 7.4636 | 9.2463  | 11.1600 | 13.7506 | 27.8002 | 50.4242 | 58.6673 | 66.3847 | 76.3675 |
| $m = 2, p = 3$ |        |         |         |         |         |         |         |         |         |
|                | 1.5755 | 2.2928  | 3.1753  | 4.6412  | 14.5958 | 33.8226 | 41.2216 | 48.3740 | 57.0102 |
| c              | 2.8737 | 3.9536  | 5.2410  | 7.0984  | 18.7054 | 39.6763 | 47.7021 | 54.9207 | 63.8283 |
| c,t            | 7.7986 | 9.7012  | 11.6899 | 14.4939 | 29.4515 | 53.3632 | 61.8438 | 69.8548 | 79.9388 |
| $m = 2, p = 4$ |        |         |         |         |         |         |         |         |         |
|                | 1.7339 | 2.4981  | 3.4703  | 5.0159  | 15.5252 | 35.8240 | 43.6428 | 51.0464 | 59.9510 |
| c              | 2.9467 | 4.0430  | 5.3852  | 7.3096  | 19.4543 | 41.4571 | 49.6628 | 57.2963 | 66.6961 |
| c,t            | 8.0506 | 10.0551 | 12.0966 | 15.0368 | 30.6521 | 55.3904 | 64.3841 | 72.5786 | 82.6272 |
| $m = 3, p = 1$ |        |         |         |         |         |         |         |         |         |
|                | 1.8991 | 2.8087  | 3.9073  | 5.4788  | 15.2663 | 33.3544 | 40.3213 | 47.0390 | 55.6948 |
| c              | 3.6828 | 4.9143  | 6.2923  | 8.2622  | 19.6794 | 39.5247 | 46.8920 | 53.8427 | 63.1706 |
| c,t            | 7.8556 | 9.7839  | 11.7727 | 14.5467 | 28.8012 | 51.7162 | 59.8743 | 67.5852 | 77.2638 |

Table 7 continued

|                | 0.010   | 0.025   | 0.050   | 0.100   | 0.500   | 0.900   | 0.950   | 0.975   | 0.990   |
|----------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| $m = 3, p = 2$ |         |         |         |         |         |         |         |         |         |
|                | 2.2619  | 3.3407  | 4.5744  | 6.3891  | 17.5405 | 37.3604 | 44.7409 | 51.6503 | 60.9317 |
| c              | 3.9874  | 5.3412  | 6.8811  | 9.1055  | 21.7137 | 43.0863 | 50.9164 | 58.1100 | 67.8342 |
| c,t            | 8.8561  | 11.0187 | 13.1809 | 16.2311 | 31.6690 | 55.6368 | 64.2565 | 72.2536 | 82.4774 |
| $m = 3, p = 3$ |         |         |         |         |         |         |         |         |         |
|                | 2.5320  | 3.7201  | 5.0262  | 6.9525  | 18.9865 | 40.0945 | 47.8203 | 55.2260 | 64.8798 |
| c              | 4.1438  | 5.5758  | 7.1904  | 9.5542  | 23.0131 | 45.5700 | 53.6396 | 61.3098 | 71.1478 |
| c,t            | 9.2529  | 11.5671 | 13.9194 | 17.1643 | 33.4200 | 58.4419 | 67.3272 | 75.5342 | 86.6035 |
| $m = 3, p = 4$ |         |         |         |         |         |         |         |         |         |
|                | 2.7677  | 4.0083  | 5.3600  | 7.3944  | 20.0353 | 42.1077 | 50.3066 | 57.8614 | 67.7364 |
| c              | 4.2703  | 5.7288  | 7.3922  | 9.8555  | 23.9210 | 47.4662 | 55.8692 | 63.8357 | 73.9377 |
| c,t            | 9.5352  | 11.9249 | 14.4225 | 17.7965 | 34.7057 | 60.6408 | 69.7676 | 78.1590 | 89.6387 |
| $m = 4, p = 1$ |         |         |         |         |         |         |         |         |         |
|                | 3.0653  | 4.3193  | 5.7271  | 7.8107  | 19.1964 | 39.0550 | 46.5991 | 53.6258 | 62.5284 |
| c              | 4.9186  | 6.4937  | 8.2245  | 10.6248 | 23.5729 | 45.1669 | 52.9950 | 60.2570 | 69.9523 |
| c,t            | 9.1915  | 11.4011 | 13.7004 | 16.8383 | 32.4819 | 56.8046 | 65.3240 | 73.2446 | 83.5502 |
| $m = 4, p = 2$ |         |         |         |         |         |         |         |         |         |
|                | 3.5018  | 4.9869  | 6.6427  | 8.9719  | 21.6929 | 43.1144 | 51.0096 | 58.7039 | 68.0398 |
| c              | 5.3188  | 7.1237  | 9.0606  | 11.6876 | 25.8405 | 48.7391 | 56.9244 | 64.9450 | 74.3894 |
| c,t            | 10.3519 | 12.7393 | 15.2253 | 18.6448 | 35.3487 | 60.8919 | 69.8928 | 78.2459 | 89.0233 |
| $m = 4, p = 3$ |         |         |         |         |         |         |         |         |         |
|                | 3.8165  | 5.4458  | 7.1799  | 9.6354  | 23.3204 | 45.9577 | 54.1944 | 61.9963 | 71.7996 |
| c              | 5.5579  | 7.4697  | 9.4988  | 12.2695 | 27.3318 | 51.3781 | 59.9915 | 68.0369 | 78.3567 |
| c,t            | 10.8577 | 13.4067 | 16.0711 | 19.6715 | 37.2752 | 63.9113 | 73.0392 | 81.6191 | 92.4438 |
| $m = 4, p = 4$ |         |         |         |         |         |         |         |         |         |
|                | 4.1029  | 5.7503  | 7.5856  | 10.1474 | 24.4741 | 48.0333 | 56.6295 | 64.6451 | 74.7573 |
| c              | 5.6837  | 7.6545  | 9.7979  | 12.6560 | 28.3573 | 53.3702 | 62.1743 | 70.4037 | 81.1153 |
| c,t            | 11.2134 | 13.8634 | 16.6048 | 20.3738 | 38.6532 | 66.1506 | 75.5562 | 84.4401 | 95.6502 |

See caption of Table 6 for further explanations

with the equation defining  $\tilde{Z}_t$  and  $\hat{\tau}$  and with  $\hat{\theta}$  denoting the OLS coefficient estimate in (1), which is now a spurious regression with an integrated error process  $\{u_t\}$ . The fact that we are in the spurious regression case implies that:

$$T^{-1}G^{-1}\hat{\theta} = \left( \sum_{t=1}^T GZ_tZ_t'G \right)^{-1} \left( \frac{1}{T} \sum_{t=1}^T T^{1/2}GZ_t \frac{y_t}{\sqrt{T}} \right) \quad (18)$$

$$\Rightarrow \left( \int_0^1 J(r)J(r)'dr \right)^{-1} \int_0^1 J(r)B_w(r)dr, \quad (19)$$

using again the notation  $\Delta y_t = w_t$  and  $\frac{1}{\sqrt{T}} y_{[rT]} = \frac{1}{\sqrt{T}} \sum_{i=1}^{[rT]} w_i \Rightarrow B_w(r)$ . This implies that  $\frac{1}{T^2} \sum_{i=1}^T \hat{u}_i^2 = \frac{1}{T^2} \hat{\tau}' \tilde{Z}_i \tilde{Z}_i' \hat{\tau} =$

$$(1, -\hat{\theta}') \begin{pmatrix} 1 & 0 \\ 0 & T^{-1} G^{-1} \end{pmatrix} \left\{ \sum_{i=1}^T \begin{pmatrix} T^{-1} & 0 \\ 0 & G \end{pmatrix} \tilde{Z}_i \tilde{Z}_i' \begin{pmatrix} T^{-1} & 0 \\ 0 & G \end{pmatrix} \right\} \begin{pmatrix} 1 & 0 \\ 0 & T^{-1} G^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ -\hat{\theta} \end{pmatrix} \quad (20)$$

converges to:

$$\tau' \int_0^1 J^*(r) J^*(r)' dr \tau, \quad (21)$$

with  $\tau$  and  $J^*(r)$  as defined in the main text. Consistency of the numerator, i.e., of  $\tilde{\Omega}_{w \cdot v}$  follows as in the proof of Phillips and Ouliaris (1990, Theorem 4.1), which relies upon the consistency of  $\hat{\xi}_t \rightarrow \xi_t$  and thus of  $\tilde{\Omega} \rightarrow \Omega$  under usual assumptions on kernel and bandwidth.  $\square$

**Proof of Proposition 3** The main issue is the existence of a bijective transformation between  $\mathbf{B}_v(r)$  and a vector of functions of standard Brownian motions. Denote with  $\tilde{\mathbf{B}}_v(r) = [B_1(r), \dots, B_m(r)]'$  and let  $\Omega_{vv}^{1/2}$  be an upper triangular matrix such that  $\tilde{\mathbf{B}}_v(r) = \Omega_{vv}^{1/2} \mathbf{W}(r)$ , with  $\mathbf{W}(r)$  a vector of standard Brownian motions. Furthermore, denote the  $(m, m)$  element of  $\Omega_{vv}$  by  $\Omega_{vv}(m, m)$ . This yields:

$$\mathbf{B}_v(r) = \begin{pmatrix} \tilde{\mathbf{B}}_v(r) \\ B_m^2(r) \\ \vdots \\ B_m^{p_m}(r) \end{pmatrix} = \begin{bmatrix} \Omega_{vv}^{1/2} & 0 & \dots & 0 \\ 0 & \Omega_{vv}(m, m) & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & \Omega_{vv}(m, m)^{p_m/2} \end{bmatrix} \begin{pmatrix} \mathbf{W}(r) \\ W_m^2(r) \\ \vdots \\ W_m^{p_m}(r) \end{pmatrix}.$$

Inserting this expression into (9) and multiplying the terms then leads to (13) and thus to the result for the CT test.

The argument for the  $\hat{P}_u$  also rests upon the above transformation with two differences: First, the considered vector is  $J^*(r)$ , i.e., also the first component corresponding to  $y_t$  is considered. Second, the block structure of  $\tau$ , with the first element equal to one, is used to simplify the limit of  $\hat{P}_u$  from (10) to the expression (14) involving only (functions) of standard Brownian motions. The calculations are analogous, with the difference being the terms with powers (and also the deterministic components), to the calculations in the proofs of Phillips and Ouliaris 1990, Lemma 2.2 and Theorem 4.1).  $\square$

**Table 8** Test results for CO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |               |                | Cubic        |                |                 |
|-------------|--------------|---------------|----------------|--------------|----------------|-----------------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$         | $CT$         | $\hat{P}_u$    | $PO_t$          |
| Australia   | 0.068        | 15.079        | −2.670         | 0.068        | 15.095         | −2.606          |
| Austria     | 0.058        | <b>69.253</b> | − <b>4.740</b> | 0.049        | <b>70.224</b>  | − <b>4.828</b>  |
| Belgium     | 0.048        | <b>96.424</b> | − <b>6.085</b> | 0.047        | <b>104.213</b> | − <b>6.677</b>  |
| Canada      | 0.090        | 13.250        | −3.340         | 0.050        | 28.482         | − <b>4.682</b>  |
| Denmark     | 0.045        | 38.912        | − <b>4.740</b> | 0.047        | 38.915         | − <b>4.736</b>  |
| Finland     | 0.041        | <b>96.490</b> | − <b>5.805</b> | 0.033        | <b>106.786</b> | − <b>6.493</b>  |
| France      | 0.055        | 45.066        | − <b>4.931</b> | 0.053        | 45.361         | − <b>4.863</b>  |
| Germany     | <b>0.123</b> | <b>67.160</b> | − <b>7.285</b> | <b>0.104</b> | <b>67.244</b>  | − <b>7.365</b>  |
| Italy       | 0.095        | 30.320        | − <b>4.445</b> | 0.078        | 45.018         | − <b>5.523</b>  |
| Japan       | 0.094        | 9.590         | − <b>5.987</b> | 0.057        | 14.294         | − <b>5.994</b>  |
| New Zealand | 0.074        | 12.087        | − <b>5.588</b> | 0.071        | 12.533         | − <b>5.965</b>  |
| Norway      | 0.074        | 28.158        | −3.381         | 0.073        | 35.655         | −3.612          |
| Portugal    | <b>0.122</b> | 18.278        | − <b>9.863</b> | <b>0.127</b> | 18.635         | − <b>10.279</b> |
| Spain       | 0.072        | 38.083        | −3.448         | 0.068        | 38.156         | −3.496          |
| Sweden      | 0.069        | 35.247        | − <b>4.180</b> | 0.071        | 36.260         | −4.262          |
| Switzerland | 0.097        | <b>85.899</b> | − <b>6.242</b> | 0.057        | <b>106.725</b> | − <b>6.769</b>  |
| UK          | 0.072        | <b>83.345</b> | − <b>6.449</b> | 0.069        | <b>83.569</b>  | − <b>6.387</b>  |
| USA         | 0.077        | 8.302         | −2.456         | 0.059        | 15.479         | −3.428          |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Bartlett kernel with bandwidth chosen according to Andrews (1991)

## Appendix B: Tables with critical values

## Supplementary Appendix C: Robustness of results

### Robustness: Kernel and bandwidth

**Table 9** Test results for CO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |               |        | Cubic        |               |        |
|-------------|--------------|---------------|--------|--------------|---------------|--------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$ | $CT$         | $\hat{P}_u$   | $PO_t$ |
| Australia   | <b>0.108</b> | 10.022        | −2.634 | <b>0.108</b> | 10.033        | −2.546 |
| Austria     | 0.050        | 50.805        | −3.797 | 0.037        | 51.517        | −3.800 |
| Belgium     | 0.059        | 36.581        | −5.758 | 0.056        | 39.536        | −6.020 |
| Canada      | <b>0.144</b> | 9.496         | −3.404 | 0.053        | 20.413        | −4.962 |
| Denmark     | 0.043        | 26.667        | −4.667 | 0.045        | 26.669        | −4.661 |
| Finland     | 0.046        | <b>64.688</b> | −5.618 | 0.030        | <b>71.591</b> | −6.090 |
| France      | 0.066        | 22.267        | −4.911 | 0.060        | 22.412        | −4.798 |
| Germany     | 0.105        | <b>57.400</b> | −8.129 | 0.087        | <b>57.472</b> | −8.218 |
| Italy       | <b>0.137</b> | 31.127        | −4.004 | 0.085        | 46.217        | −5.245 |
| Japan       | <b>0.148</b> | 8.742         | −5.828 | 0.062        | 13.031        | −5.999 |
| New Zealand | <b>0.111</b> | 14.237        | −5.419 | 0.097        | 14.762        | −5.814 |
| Norway      | <b>0.116</b> | 19.163        | −3.285 | 0.090        | 24.266        | −3.646 |

**Table 9** continued

|             | Quadratic    |               |        | Cubic        |                |        |
|-------------|--------------|---------------|--------|--------------|----------------|--------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$ | $CT$         | $\hat{P}_u$    | $PO_t$ |
| Portugal    | <b>0.106</b> | 20.157        | −9.129 | <b>0.110</b> | 20.550         | −9.364 |
| Spain       | 0.088        | 35.162        | −3.380 | 0.087        | 35.229         | −3.466 |
| Sweden      | 0.082        | 25.377        | −4.403 | 0.082        | 26.106         | −4.469 |
| Switzerland | 0.091        | <b>86.834</b> | −6.078 | 0.050        | <b>107.887</b> | −6.657 |
| UK          | 0.066        | <b>92.738</b> | −7.055 | 0.063        | <b>92.987</b>  | −7.004 |
| USA         | <b>0.152</b> | 6.582         | −2.395 | 0.075        | 12.272         | −3.659 |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Quadratic Spectral kernel with bandwidth chosen according to Newey and West (1994)

**Table 10** Test results for CO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |               |                | Cubic        |                |                 |
|-------------|--------------|---------------|----------------|--------------|----------------|-----------------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$         | $CT$         | $\hat{P}_u$    | $PO_t$          |
| Australia   | 0.083        | 14.723        | −2.670         | 0.086        | 14.739         | −2.609          |
| Austria     | 0.058        | <b>75.487</b> | − <b>4.846</b> | 0.050        | <b>76.545</b>  | − <b>4.933</b>  |
| Belgium     | 0.040        | <b>98.242</b> | − <b>6.135</b> | 0.044        | <b>106.178</b> | − <b>6.718</b>  |
| Canada      | 0.076        | 10.237        | −3.339         | 0.043        | 22.005         | − <b>4.614</b>  |
| Denmark     | 0.038        | 38.968        | − <b>4.726</b> | 0.041        | 38.971         | − <b>4.721</b>  |
| Finland     | 0.035        | <b>95.854</b> | − <b>5.874</b> | 0.030        | <b>106.083</b> | − <b>6.558</b>  |
| France      | 0.047        | 39.043        | − <b>4.929</b> | 0.045        | 39.298         | − <b>4.865</b>  |
| Germany     | <b>0.116</b> | <b>58.498</b> | − <b>7.247</b> | 0.097        | <b>58.571</b>  | − <b>7.328</b>  |
| Italy       | 0.081        | 25.955        | − <b>4.484</b> | 0.067        | 38.537         | − <b>5.603</b>  |
| Japan       | 0.078        | 8.913         | − <b>5.960</b> | 0.050        | 13.286         | − <b>5.997</b>  |
| New Zealand | 0.073        | 11.792        | − <b>5.549</b> | 0.069        | 12.227         | − <b>5.932</b>  |
| Norway      | 0.060        | 26.703        | −3.383         | 0.062        | 33.813         | −3.609          |
| Portugal    | <b>0.113</b> | 16.698        | − <b>9.870</b> | <b>0.119</b> | 17.024         | − <b>10.277</b> |
| Spain       | 0.063        | 35.456        | −3.445         | 0.058        | 35.524         | −3.499          |
| Sweden      | 0.060        | 35.546        | −4.150         | 0.062        | 36.568         | −4.220          |
| Switzerland | 0.090        | <b>87.613</b> | − <b>6.213</b> | 0.051        | <b>108.855</b> | − <b>6.751</b>  |
| UK          | 0.064        | <b>73.503</b> | − <b>6.377</b> | 0.061        | <b>73.700</b>  | − <b>6.316</b>  |
| USA         | 0.062        | 7.072         | −2.266         | 0.051        | 13.187         | −3.108          |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Quadratic Spectral kernel with bandwidth chosen according to Andrews (1991)



**Table 11** Test results for SO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |             |               | Cubic        |             |               |
|-------------|--------------|-------------|---------------|--------------|-------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$        | $CT$         | $\hat{P}_u$ | $PO_t$        |
| Australia   | 0.050        | 38.851      | −3.275        | 0.047        | 40.471      | −3.384        |
| Austria     | 0.072        | 28.834      | <b>−4.191</b> | 0.051        | 39.378      | −4.073        |
| Belgium     | 0.081        | 19.571      | −4.037        | 0.043        | 30.249      | −3.776        |
| Canada      | 0.055        | 25.583      | −3.599        | 0.053        | 25.741      | −3.649        |
| Denmark     | 0.061        | 11.836      | −3.244        | 0.045        | 24.701      | −4.009        |
| Finland     | 0.063        | 28.260      | −4.100        | 0.065        | 28.717      | −3.910        |
| France      | 0.095        | 6.345       | −2.346        | 0.096        | 13.468      | −2.255        |
| Germany     | 0.082        | 5.941       | −2.665        | 0.074        | 11.303      | −2.590        |
| Italy       | <b>0.109</b> | 3.572       | −2.276        | <b>0.209</b> | 3.810       | −1.559        |
| Japan       | 0.084        | 15.155      | −2.957        | 0.049        | 19.087      | −4.118        |
| New Zealand | 0.072        | 9.859       | <b>−4.862</b> | 0.049        | 13.441      | <b>−7.054</b> |
| Norway      | 0.064        | 26.826      | −3.155        | 0.066        | 27.077      | −3.231        |
| Portugal    | 0.074        | 13.692      | −2.968        | 0.076        | 13.752      | −3.073        |
| Spain       | <b>0.117</b> | 12.100      | −1.699        | <b>0.163</b> | 13.089      | −1.487        |
| Sweden      | 0.053        | 13.118      | −3.436        | 0.058        | 14.438      | −2.967        |
| Switzerland | 0.105        | 9.960       | −2.217        | 0.060        | 55.467      | <b>−5.056</b> |
| UK          | 0.087        | 10.208      | −1.629        | 0.063        | 21.014      | −2.703        |
| USA         | <b>0.218</b> | 9.985       | −0.826        | <b>0.219</b> | 10.039      | −0.811        |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Bartlett kernel with bandwidth chosen according to Andrews (1991)

**Table 12** Test results for SO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |             |               | Cubic |             |               |
|-------------|--------------|-------------|---------------|-------|-------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$        | $CT$  | $\hat{P}_u$ | $PO_t$        |
| Australia   | 0.049        | 37.798      | −3.399        | 0.045 | 39.374      | −3.519        |
| Austria     | 0.075        | 25.453      | −3.633        | 0.052 | 34.759      | −3.608        |
| Belgium     | 0.096        | 15.796      | −3.701        | 0.040 | 24.414      | −3.872        |
| Canada      | 0.051        | 21.662      | −3.994        | 0.048 | 21.796      | −4.018        |
| Denmark     | 0.098        | 9.939       | −3.203        | 0.054 | 20.743      | −4.036        |
| Finland     | 0.076        | 23.523      | −3.977        | 0.073 | 23.903      | −3.905        |
| France      | <b>0.107</b> | 5.214       | −2.342        | 0.063 | 11.069      | −2.566        |
| Germany     | 0.096        | 5.821       | −2.534        | 0.060 | 11.075      | −2.861        |
| Italy       | 0.085        | 6.081       | −2.299        | 0.086 | 6.486       | −1.809        |
| Japan       | <b>0.111</b> | 15.898      | −3.284        | 0.031 | 20.022      | <b>−4.534</b> |
| New Zealand | 0.089        | 12.421      | <b>−4.867</b> | 0.051 | 16.934      | <b>−6.968</b> |

**Table 12** continued

|             | Quadratic    |             |        | Cubic |             |               |
|-------------|--------------|-------------|--------|-------|-------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$ | $CT$  | $\hat{P}_u$ | $PO_t$        |
| Norway      | 0.066        | 21.564      | -3.220 | 0.064 | 21.766      | -3.276        |
| Portugal    | 0.099        | 11.272      | -3.207 | 0.100 | 11.322      | -3.303        |
| Spain       | 0.071        | 11.803      | -1.865 | 0.077 | 12.768      | -1.857        |
| Sweden      | 0.063        | 10.275      | -3.364 | 0.065 | 11.310      | -2.907        |
| Switzerland | <b>0.207</b> | 5.453       | -1.655 | 0.055 | 30.369      | <b>-5.156</b> |
| UK          | 0.072        | 2.175       | -1.723 | 0.060 | 4.477       | -2.747        |
| USA         | 0.081        | 7.554       | -1.351 | 0.071 | 7.595       | -1.324        |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Quadratic Spectral kernel with bandwidth chosen according to Newey and West (1994)

**Table 13** Test results for SO<sub>2</sub> emissions for the sample range 1870–2016 (for New Zealand 1878–2016)

|             | Quadratic    |             |               | Cubic        |               |               |
|-------------|--------------|-------------|---------------|--------------|---------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$        | $CT$         | $\hat{P}_u$   | $PO_t$        |
| Australia   | 0.056        | 38.767      | -3.276        | 0.054        | 40.383        | -3.385        |
| Austria     | 0.065        | 31.234      | <b>-4.260</b> | 0.047        | 42.655        | -4.135        |
| Belgium     | 0.072        | 17.997      | -4.085        | 0.046        | 27.816        | -3.757        |
| Canada      | 0.050        | 21.331      | -3.567        | 0.049        | 21.463        | -3.606        |
| Denmark     | 0.050        | 11.538      | -3.240        | 0.040        | 24.078        | -4.055        |
| Finland     | 0.056        | 28.437      | <b>-4.168</b> | 0.059        | 28.897        | -3.972        |
| France      | 0.100        | 5.998       | -2.370        | <b>0.266</b> | 12.732        | -2.364        |
| Germany     | 0.087        | 5.983       | -2.650        | <b>0.110</b> | 11.383        | -2.611        |
| Italy       | <b>0.176</b> | 2.954       | -2.309        | <b>0.953</b> | 3.151         | -1.554        |
| Japan       | 0.073        | 15.490      | -2.907        | 0.057        | 19.508        | -4.048        |
| New Zealand | 0.090        | 9.577       | <b>-4.853</b> | 0.048        | 13.056        | <b>-7.041</b> |
| Norway      | 0.066        | 27.005      | -3.185        | 0.068        | 27.257        | -3.242        |
| Portugal    | 0.078        | 13.992      | -2.962        | 0.081        | 14.053        | -3.075        |
| Spain       | <b>0.339</b> | 12.107      | -1.733        | <b>0.768</b> | 13.097        | -1.551        |
| Sweden      | 0.046        | 11.847      | -3.292        | 0.059        | 13.040        | -2.848        |
| Switzerland | 0.097        | 10.241      | -2.176        | 0.058        | <b>57.034</b> | <b>-5.106</b> |
| UK          | <b>0.161</b> | 10.714      | -1.758        | 0.066        | 22.056        | -2.660        |
| USA         | <b>6.276</b> | 8.172       | -0.801        | <b>8.516</b> | 8.216         | -0.766        |

For further explanations see caption of Table 2 with the difference that the results in this table are based on the Quadratic Spectral kernel with bandwidth chosen according to Andrews (1991)

**Robustness: Sample range until 2000****Table 14** Test results for CO<sub>2</sub> emissions for the sample range 1870–2000 (for New Zealand 1878–2000)

|             | Quadratic    |               |                       | Cubic        |                |                       |
|-------------|--------------|---------------|-----------------------|--------------|----------------|-----------------------|
|             | <i>CT</i>    | $\hat{P}_u$   | <i>PO<sub>t</sub></i> | <i>CT</i>    | $\hat{P}_u$    | <i>PO<sub>t</sub></i> |
| Australia   | <b>0.107</b> | 11.294        | −2.623                | <b>0.106</b> | 11.306         | −2.530                |
| Austria     | 0.055        | <b>57.186</b> | −3.816                | 0.042        | <b>57.988</b>  | −3.822                |
| Belgium     | 0.061        | <i>51.074</i> | − <b>5.710</b>        | 0.057        | <i>55.199</i>  | − <b>5.667</b>        |
| Canada      | <b>0.144</b> | 11.882        | −3.355                | 0.056        | 25.542         | − <b>4.878</b>        |
| Denmark     | 0.047        | 36.027        | − <b>4.673</b>        | 0.049        | 36.029         | − <b>4.670</b>        |
| Finland     | 0.048        | <b>75.487</b> | − <b>5.719</b>        | 0.033        | <b>83.542</b>  | − <b>6.136</b>        |
| France      | 0.066        | 27.859        | − <b>4.929</b>        | 0.062        | 28.041         | − <b>4.863</b>        |
| Germany     | <b>0.108</b> | <b>69.262</b> | − <b>8.001</b>        | <i>0.090</i> | <b>69.349</b>  | − <b>8.089</b>        |
| Italy       | <b>0.138</b> | 34.120        | − <b>4.174</b>        | <i>0.088</i> | <i>50.661</i>  | − <b>5.520</b>        |
| Japan       | <b>0.149</b> | 8.718         | − <b>5.889</b>        | 0.065        | 12.995         | − <b>6.009</b>        |
| New Zealand | <b>0.112</b> | 13.698        | − <b>5.422</b>        | <i>0.098</i> | 14.203         | − <b>5.809</b>        |
| Norway      | <b>0.116</b> | 20.583        | −3.334                | <i>0.092</i> | 26.063         | −3.617                |
| Portugal    | <b>0.108</b> | 20.919        | − <b>9.181</b>        | <b>0.111</b> | 21.327         | − <b>9.412</b>        |
| Spain       | <i>0.091</i> | 42.031        | −3.315                | <i>0.089</i> | 42.112         | −3.384                |
| Sweden      | 0.084        | 29.150        | − <b>4.312</b>        | <i>0.084</i> | 29.988         | −4.323                |
| Switzerland | <i>0.097</i> | <b>86.476</b> | − <b>6.291</b>        | 0.056        | <b>107.442</b> | − <b>6.763</b>        |
| UK          | 0.070        | <b>91.459</b> | − <b>6.894</b>        | 0.067        | <b>91.704</b>  | − <b>6.848</b>        |
| USA         | <b>0.148</b> | 12.620        | −2.337                | 0.077        | 23.531         | −3.558                |

For further explanations see caption of Table 2

**Table 15** Test results for SO<sub>2</sub> emissions for the sample range 1870–2000 (for New Zealand 1878–2000)

|           | Quadratic    |             |                       | Cubic     |             |                       |
|-----------|--------------|-------------|-----------------------|-----------|-------------|-----------------------|
|           | <i>CT</i>    | $\hat{P}_u$ | <i>PO<sub>t</sub></i> | <i>CT</i> | $\hat{P}_u$ | <i>PO<sub>t</sub></i> |
| Australia | 0.051        | 38.885      | −3.402                | 0.048     | 40.506      | −3.521                |
| Austria   | 0.080        | 28.695      | −3.776                | 0.055     | 39.188      | −3.695                |
| Belgium   | <i>0.100</i> | 17.752      | −3.738                | 0.043     | 27.436      | −3.556                |
| Canada    | 0.055        | 22.976      | −3.865                | 0.053     | 23.118      | −3.888                |
| Denmark   | <i>0.101</i> | 10.010      | −3.160                | 0.057     | 20.890      | −4.006                |
| Finland   | 0.078        | 25.292      | −4.082                | 0.077     | 25.701      | −3.905                |
| France    | <b>0.112</b> | 5.524       | −2.353                | 0.067     | 11.725      | −2.455                |
| Germany   | <i>0.098</i> | 6.057       | −2.584                | 0.062     | 11.524      | −2.782                |

**Table 15** continued

|             | Quadratic    |             |               | Cubic        |             |               |
|-------------|--------------|-------------|---------------|--------------|-------------|---------------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$        | $CT$         | $\hat{P}_u$ | $PO_t$        |
| Italy       | 0.089        | 3.835       | -2.296        | 0.090        | 4.090       | -1.729        |
| Japan       | <b>0.113</b> | 15.237      | -3.219        | 0.036        | 19.190      | -4.447        |
| New Zealand | 0.090        | 11.831      | <b>-4.849</b> | 0.054        | 16.130      | <b>-6.941</b> |
| Norway      | 0.069        | 22.719      | -3.216        | 0.067        | 22.931      | -3.283        |
| Portugal    | 0.100        | 11.880      | -3.175        | <b>0.102</b> | 11.933      | -3.271        |
| Spain       | 0.074        | 12.060      | -1.799        | 0.080        | 13.046      | -1.761        |
| Sweden      | 0.066        | 11.155      | -3.709        | 0.067        | 12.278      | -2.908        |
| Switzerland | <b>0.203</b> | 6.766       | -1.703        | 0.060        | 37.684      | <b>-5.067</b> |
| UK          | 0.075        | 0.841       | -1.510        | 0.066        | 1.731       | -2.647        |
| USA         | 0.085        | 10.517      | -1.230        | 0.074        | 10.574      | -1.200        |

For further explanations see caption of Table 2

### Robustness: Old data vintage (sample range until 2000)

**Table 16** Test results for CO<sub>2</sub> emissions for the sample range 1870–2000 (for New Zealand 1878–2000) using the earlier data vintage

|             | Quadratic    |               |               | Cubic        |               |               |
|-------------|--------------|---------------|---------------|--------------|---------------|---------------|
|             | $CT$         | $\hat{P}_u$   | $PO_t$        | $CT$         | $\hat{P}_u$   | $PO_t$        |
| Australia   | <b>0.296</b> | 11.560        | -2.886        | <b>0.259</b> | 12.868        | -2.498        |
| Austria     | 0.083        | <b>63.516</b> | <b>-4.504</b> | 0.083        | <b>64.114</b> | <b>-4.547</b> |
| Belgium     | 0.095        | <b>66.359</b> | <b>-5.661</b> | 0.069        | <b>68.694</b> | <b>-5.994</b> |
| Canada      | <b>0.286</b> | 11.110        | -3.231        | 0.098        | 25.475        | <b>-4.776</b> |
| Denmark     | <b>0.119</b> | <b>49.891</b> | <b>-4.901</b> | 0.100        | 51.498        | <b>-4.943</b> |
| Finland     | 0.075        | <b>73.254</b> | <b>-5.390</b> | 0.042        | <b>79.318</b> | <b>-5.906</b> |
| France      | <b>0.107</b> | 36.151        | <b>-4.822</b> | <b>0.113</b> | 36.193        | <b>-4.857</b> |
| Germany     | <b>0.257</b> | <b>61.717</b> | <b>-6.699</b> | <b>0.212</b> | <b>66.902</b> | <b>-7.610</b> |
| Italy       | <b>0.158</b> | 39.011        | -3.952        | <b>0.164</b> | 39.024        | -3.952        |
| Japan       | <b>0.293</b> | 6.635         | <b>-5.629</b> | <b>0.130</b> | 10.144        | <b>-5.650</b> |
| New Zealand | <b>0.270</b> | 11.111        | <b>-5.198</b> | <b>0.132</b> | 13.851        | <b>-7.175</b> |
| Norway      | <b>0.191</b> | 25.439        | <b>-4.080</b> | <b>0.164</b> | 29.202        | -3.516        |
| Portugal    | <b>0.222</b> | 18.394        | <b>-9.126</b> | <b>0.207</b> | 18.922        | <b>-9.537</b> |

**Table 16** continued

|             | Quadratic    |                |        | Cubic        |                |        |
|-------------|--------------|----------------|--------|--------------|----------------|--------|
|             | $CT$         | $\hat{P}_u$    | $PO_t$ | $CT$         | $\hat{P}_u$    | $PO_t$ |
| Spain       | <b>0.183</b> | 36.268         | -3.449 | <b>0.169</b> | 36.768         | -3.604 |
| Sweden      | <b>0.170</b> | 29.349         | -3.604 | <b>0.131</b> | 32.206         | -3.666 |
| Switzerland | <b>0.268</b> | 31.297         | -3.764 | <b>0.105</b> | 46.214         | -5.357 |
| UK          | <b>0.125</b> | <b>130.958</b> | -8.997 | <b>0.111</b> | <b>136.550</b> | -9.118 |
| USA         | <b>0.357</b> | 12.448         | -2.151 | <b>0.152</b> | 25.426         | -3.869 |

For further explanations see caption of Table 2

**Table 17** Test results for SO<sub>2</sub> emissions for the sample range 1870–2000 (for New Zealand 1878–2000) using the earlier data vintage

|             | Quadratic    |             |        | Cubic        |               |        |
|-------------|--------------|-------------|--------|--------------|---------------|--------|
|             | $CT$         | $\hat{P}_u$ | $PO_t$ | $CT$         | $\hat{P}_u$   | $PO_t$ |
| Australia   | <b>0.215</b> | 20.375      | -2.746 | <b>0.214</b> | 20.404        | -2.708 |
| Austria     | <b>0.165</b> | 15.311      | -3.478 | <b>0.175</b> | 20.231        | -3.645 |
| Belgium     | <b>0.188</b> | 26.919      | -4.068 | 0.088        | 42.594        | -4.129 |
| Canada      | 0.074        | 21.233      | -4.879 | 0.068        | 21.362        | -4.949 |
| Denmark     | <b>0.266</b> | 24.030      | -1.577 | <b>0.153</b> | 44.356        | -3.758 |
| Finland     | <b>0.120</b> | 17.851      | -2.967 | <b>0.131</b> | 19.342        | -2.982 |
| France      | <b>0.159</b> | 11.392      | -2.410 | <b>0.107</b> | 18.982        | -3.186 |
| Germany     | <b>0.131</b> | 17.167      | -2.237 | <b>0.121</b> | 21.299        | -2.672 |
| Italy       | <b>0.129</b> | 23.783      | -2.554 | <b>0.147</b> | 31.723        | -3.065 |
| Japan       | <b>0.255</b> | 5.853       | -3.538 | <b>0.150</b> | 7.442         | -3.919 |
| New Zealand | <b>0.226</b> | 20.312      | -2.785 | <b>0.166</b> | 22.332        | -3.322 |
| Norway      | <b>0.151</b> | 22.654      | -3.680 | <b>0.153</b> | 22.990        | -3.765 |
| Portugal    | <b>0.217</b> | 46.728      | -4.589 | <b>0.206</b> | 47.775        | -4.611 |
| Spain       | <b>0.180</b> | 13.689      | -2.559 | <b>0.221</b> | 14.580        | -2.349 |
| Sweden      | <b>0.130</b> | 16.067      | -3.128 | <b>0.146</b> | 17.231        | -3.100 |
| Switzerland | <b>0.119</b> | 46.846      | -5.564 | <b>0.112</b> | 49.448        | -5.559 |
| UK          | <b>0.236</b> | 28.799      | -2.899 | 0.075        | <b>58.572</b> | -5.817 |
| USA         | <b>0.315</b> | 12.041      | -2.369 | <b>0.142</b> | 18.025        | -3.226 |

For further explanations see caption of Table 2

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